ROHIT RAKSHAN PALANIKUMAR

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EDUCATION

Bachelor of Science in Aerospace Engineering (Honors) – University of Texas at Austin **August 2023 - December 2026**

EXPERIENCE

Shield AI – *Junior Manufacturing Engineer* **June 2026 – August 2026**

- Fully owned forward-to-aft build and WI, maintaining tight 0.020in tolerance via CMM to prevent catastrophic failure
- Designed and rapidly implemented an assembly fixture to maintain aircraft structure critical datums within 0.050in
- Authored production and prototype work instructions, optimizing build sequencing to reduce wing assembly time by 29%
- Resolved production issues involving build sequence, tooling access, and torque specifications with cross-functional teams

The Boring Company – *Integration Engineering Intern* **September 2025 – December 2025**

- Led the assembly and integration of 3+ Prufrock Tunnel Boring Machines (TBMs) across multi-disciplinary builds
- Optimized Manufacturing Bills of Materials (MBOMs) across 5 active projects with 500+ parts and corrected design errors
- Executed hands-on mechanical assembly with power tools, torque-controlled fastening, subsystem integration, and manufacturing support alongside technicians and engineers
- Supported rapid hardware refurbishment and field-driven integration efforts under aggressive production schedules

Flamefront Propulsion – *Engineering Intern* **June 2025 – August 2025**

- Built turbojet engine modeling tool in Python from scratch using GitHub libraries, ingesting Mach, altitude, and geometry inputs to output thrust, SFC, and required propulsion performance metrics across the flight envelope
- Implemented component selection logic with mechanical and thermodynamic constraints; achieved 6% thrust increase
- Designed engine geometry in CAD from component analysis outputs, closing analysis-to-hardware loop

Computational Fluid Physics Laboratory – *Undergraduate Researcher* **October 2024 – August 2025**

- Investigated programmable microjet flow control in turbulent boundary layers to extend attached flow over NACA 4412
- Ran high-fidelity CFD cases on TACC supercomputer using Nek5000; 93% match with XFOIL Cl/Cd due to span increase
- Microjet made a 42% decrease in separation volume for fast LSMs; indicating aerodynamic performance improvement

ORGANIZATIONS

Longhorn Racing Electric (FSAE Team) – *Aerodynamics Engineer* **August 2023 – Present**

- Designed aerodynamic and structural composite components in SolidWorks including sidepods, flip-ups, and bargeboards
- Performed CFD-driven aerodynamic optimization in ANSYS and Cadence while meeting manufacturability constraints
- Performed hands-on machining using mills and lathes to fabricate precision components and tooling fixtures
- Manufactured lightweight aero components using carbon fiber layups and resin infusion, foam cores, Nomex cores, aluminum ribs and Carbon Fiber spars to improve stiffness-weight ratio under racing conditions

Texas Spacecraft Laboratory

Communications Test Engineer **February 2024 – May 2025**

- Conducted thermal-vacuum testing of spacecraft electronics across -50°C to +100°C and 10^{-3} Pa, documenting results and validating component performance under flight-representative environmental conditions

LRA Vertical Active Stabilization – *Avionics Bay Sub-Team Lead* **August 2023 – March 2024**

- Spearheaded FEA based design and build of the avionics bay to house flight electronics, ensuring structural fit and component accessibility for integration and recovery

PROJECTS

TLOFT DARPA Heavy Lift Challenge

- Designed full VTOL composite airframe (7 ft wingspan hybrid configuration) targeting 4:1 payload-to-aircraft weight ratio
- Performed hand calculations (buckling, cantilever etc.) to validate structural integrity for primary load-bearing members
- Machined, fabricated, and assembled composite airframe while utilizing 3D Printing to produce integration hardware
- Simulated VTOL trajectory via rotor dynamics modeling; optimal cruise AoA of 14° at 18 m/s for heavy-lift flight envelope

Automatic Headrest

- Designed a cost-effective off-the-shelf part that adjusts a car seat headrest using a linear actuator and switches (soldering)
- Integrated hidden wiring and bolted components onto the frame for maximum stability

SKILLS

- **Digital:** SolidWorks, NX, Inventor, ANSYS, Cadence, MATLAB, ParaView , Java, C++, Python, R, HTML, GridPro, Git, SW Dev
- **Manufacturing:** Composite Fabrication, Aircraft Assembly, Lathe, Mill, CAM, Carbon Fiber Layups, Resin Infusion, Mechanical Integration, 3D Printing, Soldering, Hand Tools